HW7

Shamim Kazi
2024-11-18

Section 4.3

4.4-1

(b)

$$f(x,y) = \frac{4xy}{9} \quad x = 1, 2 \text{ and } y = 1, 2, 3, 4$$

First get(s):

$$g(x) = \frac{4xy}{9} \quad x, y = 1, 2$$

For $x = 1$

$$g(1) = \frac{4(1)(1)}{9} = \frac{4}{9}$$
$$g(1) = \frac{4(1)(2)}{9} = \frac{8}{9}$$
$$g(1) = \frac{4(1)(3)}{9} = \frac{12}{9}$$
$$g(1) = \frac{4(1)(4)}{9} = \frac{16}{9}$$

For $x = 2$

$$g(2) = \frac{4(2)(1)}{9} = \frac{8}{9}$$
$$g(2) = \frac{4(2)(2)}{9} = \frac{16}{9}$$
$$g(2) = \frac{4(2)(3)}{9} = \frac{24}{9}$$
$$g(2) = \frac{4(2)(4)}{9} = \frac{32}{9}$$

(c)

$$h(x) = \frac{4xy}{9}$$

$$h(1) = \frac{4(1)(1)}{9} = \frac{4}{9}$$
$$h(1) = \frac{4(1)(2)}{9} = \frac{8}{9}$$
$$h(1) = \frac{4(1)(3)}{9} = \frac{12}{9}$$
$$h(1) = \frac{4(1)(4)}{9} = \frac{16}{9}$$
$$h(2) = \frac{4(2)(1)}{9} = \frac{8}{9}$$
$$h(2) = \frac{4(2)(2)}{9} = \frac{16}{9}$$
$$h(2) = \frac{4(2)(3)}{9} = \frac{24}{9}$$
$$h(2) = \frac{4(2)(4)}{9} = \frac{32}{9}$$

(d)

$$P(A \text{ or } B) = P(A) + P(B)$$
$$P(1 \le X \le 2 \text{ or } 1 \le Y \le 2) = P(1 \le X \le 2) + P(1 \le Y \le 2)$$
$$= \sum_{x=1}^{2} h(x)$$
$$P(1 \le X \le 2) = h(1) + h(2)$$
$$= \frac{12}{9} + \frac{24}{9}$$
$$P(1 \le X \le 2) = \frac{36}{9}$$

(e)

Find E(X) if $h = \text{Lower Fun}(X \text{ or } Y) =$

```
1 2 4 3 6 8 9 12 16 4 8 12 16 9 18 24 36 16 32 48 64
```

```
36 / 9 = 4.0000000
```

$$E(X) = \sum_{x=1}^{2} x \cdot g(x)$$

```
E(X) = 1 * g(1) + 2 * g(2)
```

```
1 * 4 / 9 + 2 * 8 / 9 = 4.0000000
```

$$E(X^2) = \sum_{x=1}^{2} x^2 \cdot g(x)$$

```
E(X^2) = 1^2 * g(1) + 2^2 * g(2)
```

```
1^2 * 4 / 9 + 2^2 * 8 / 9 = 7.0000000
```

$$Var(X) = E(X^2) - E(X)^2$$

```
Var(X) = E(X^2) - E(X)^2
```

```
7.0000000 - 4.0000000^2 = 0.9999999
```

4.3-2

$$(X, R) \sim Bernoulli$$
$$(X, W) \sim Bernoulli$$
$$(R, W) \sim Bernoulli$$
$$(X, W) \sim Binomial$$
$$P(Bernoulli = 1) = p, \text{ or } p = (R, W)$$
$$P(Bernoulli = 0)$$
$$X \sim Binomial(n = 100, p = \frac{1}{6})$$

(b)

```
Expected Value = n * p
= 100 * p = 16.7
```

```
n <- 100
p <- 1/6
n*p
```

```
[1] 16.67
```

```
E(X) = 100
```

```
Var(X) = n * p(1 - p)
```

```
n <- 100
p <- 1/6
n*p*(1-p)
```

```
[1] 13.89
```

```
Var(X) = 100
```

(c) Given that X = 300, how is Y distributed?

$$(X, R) \sim Binomial(n = 300, p = \frac{1}{6})$$
$$(X, W) \sim Binomial(n = 300, p = \frac{1}{6})$$
$$P(Binomial(n = 300, p = \frac{1}{6}) = 300) = 0.0000$$

$$X \mid Y = 300 \sim Binomial(300, \frac{1}{6})$$

(d) Give the values of E(Y | X = 300) and Var(Y | X = 300).

```
n <- 300
p <- 1/6
n*p
```

```
[1] 50
```

```
E(Y | X = 300) = 50
```

```
n <- 300
p <- 1/6
n*p*(1-p)
```

```
[1] 41.67
```

```
Var(Y | X = 300) = 41.67
```

4.3-6

PD-100:

```
1/.01 = 0.0001.ln
```

```
[1] 9.2
```

```
E X = 9.2
```

PD-100:

```
1/.0001.ln(2)
```

```
[1] 6.93
```

```
Var X = 6.93
```

PD = MD (0.100)

$$E X = \frac{\ln(2)}{\ln(1.100)}$$

```
10.25/10.75
```

```
[1] 0.9
```

```
Var X = 0.9
```

PD = MD (0.100)

$$Var X = \frac{\ln(2)(1 - .100)}{\ln(1.100)^2}$$

```
0.9/.1
```

```
[1] 9.00
```

```
Var X = 9.00
```

(b)

$$\alpha_0 = \sum_{x=0}^{\infty} \alpha_x = P(X = 0)$$

```
100 * 0.01 = 100 * 0.0 = 100 * 0.00
```

```
[1] 0.00
```

```
\alpha_0 = 0
```

```
1000 * 0.0 = 1000 * 0.00 = 100 * 0.00
```

```
[1] 0.00
```

```
\alpha_0 = 0
```

$$\alpha_x = \sum_{x=1}^{\infty} \alpha_x = P(X \ge x)$$

```
10.0/100.0*(1 - 0.01) = 0.0990100 * 0.01 = 0.0009900 * 0.01
```

```
[1] 1.0e-05
```

```
\alpha = 1e-05
```

$$\alpha_x = \sum_{x=1}^{\infty} \alpha_x = P(X \ge x)$$

```
1.0/99.1 = 0.0101 * 0.01 = 0.0000101 * 0.01 = 0.0000000101 * 0.01
```

```
[1] 1.0e-10
```

```
\alpha = 1e-10
```

(c) Compute the conditional means E(X | Y = 100), E(X | Y = 200).

$$E(X \mid Y = 100) = \sum_{x=0}^{\infty} x \cdot P(X = x \mid Y = 100)$$
$$\frac{\alpha_x \cdot P(Y = 100 \mid X = x)}{\sum_{x=0}^{\infty} \alpha_x \cdot P(Y = 100 \mid X = x)} = 100$$

```
(10.0/.1) * (1/.1) * 100 * 0.1 * 0.9 * 100 * 0.01
```

```
[1] 10.0000000
```

```
Cond(X, Y) = E(X^2) - E(X)E(Y)
```

```
100/90 = (10) * 100
```

```
[1] 1.0e+03
```

```
\alpha = 1e+03
```

(d) Compute the covariance coefficient ρ = Cov(X, Y)/σX σY.

$$\rho = \frac{Cov(X, Y)}{\sigma_X \cdot \sigma_Y}$$

```
sd(c(0,0,0,
```

$$2.1902$$

$$= \phi(1.23) - \phi(-1.96)$$

$$0.8914 - 0.0249 = \boxed{0.8665}$$